

# Earnings Leverage Stability and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Earnings Leverage Stability (ELS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ELS achieves an annualized gross (net) Sharpe ratio of 0.46 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 28 (21) bps/month with a t-statistic of 3.68 (2.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance) is 22 bps/month with a t-statistic of 3.06.

# 1 Introduction

Asset prices reflect investors’ marginal rates of substitution between current and future consumption, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While traditional asset pricing models focus on contemporaneous market risk, a growing literature emphasizes the importance of firms’ sensitivity to long-run consumption dynamics and state-dependent risk premia (Bansal and Yaron, 2004; Campbell and Cochrane, 1999a). Despite extensive research documenting hundreds of return predictors, we lack a comprehensive understanding of how firms’ earnings leverage stability relates to their fundamental exposure to consumption risk during different economic states.

Earnings Leverage Stability (ELS) captures the temporal consistency of a firm’s operating leverage, measuring how reliably earnings respond to changes in sales over time. From a consumption-based asset pricing perspective, firms with unstable earnings leverage face greater exposure to aggregate consumption risk through multiple channels (Cochrane and Piazzesi, 2005; Lettau and Ludvigson, 2005). First, volatile operating leverage amplifies firms’ sensitivity to long-run consumption shocks, as uncertain cost structures make cash flows more sensitive to persistent changes in aggregate demand. During economic downturns when marginal utility is high, firms with unstable earnings leverage experience disproportionate deterioration in profitability, making their equity claims particularly unattractive hedges against consumption risk (Campbell and Cochrane, 1999b). Second, earnings leverage instability increases firms’ exposure to disaster risk and rare consumption disasters. When facing tail events that sharply reduce aggregate consumption, firms with volatile operating leverage cannot reliably adjust their cost base, leading to severe cash flow impairments (Barro, 2006; Gabaix, 2012). This disaster sensitivity manifests through both direct channels—as consumption disasters reduce revenues—and indirect channels through disrupted supply chains and credit markets. Third, the state-dependence of

risk premia implies that earnings leverage stability matters most during bad times when the intertemporal marginal rate of substitution is high. Firms with stable earnings leverage provide valuable consumption hedging properties by maintaining predictable cash flows when investors most value them, while those with unstable leverage amplify consumption risk precisely when risk aversion peaks (Campbell and Vuolteenaho, 2004; Bansal et al., 2012).

We find strong evidence that Earnings Leverage Stability predicts cross-sectional variation in stock returns. A value-weighted long/short trading strategy based on ELS achieves an annualized gross Sharpe ratio of 0.46, with a monthly average return of 25 basis points (t-statistic = 3.33). The strategy’s alpha remains economically and statistically significant across all standard factor models, ranging from 27 to 30 basis points per month with t-statistics exceeding 3.66. Notably, the strategy loads negatively on the market factor ( $\beta = -0.06$ , t-statistic = -3.49), consistent with ELS capturing countercyclical risk premia that increase during periods of high marginal utility. The economic significance of our findings persists after accounting for transaction costs. The net Sharpe ratio of 0.34 places the ELS strategy in the top 7% of documented anomalies, while the strategy’s net generalized alpha relative to the Fama-French six-factor model equals 21 basis points per month (t-statistic = 2.89). These results demonstrate that ELS captures a robust source of priced consumption risk not subsumed by existing factors. The predictive power of ELS extends across the size spectrum, with the strategy generating significant returns even among the largest stocks. Within the top NYSE size quintile, the ELS strategy earns 25 basis points per month (t-statistic = 2.86) with alphas ranging from 23 to 31 basis points relative to standard factor models. This pervasive pricing across market capitalizations suggests that ELS captures systematic exposure to aggregate consumption risk rather than microstructure effects or limits to arbitrage.

Our study contributes to the consumption-based asset pricing literature by iden-

tifying earnings leverage stability as a novel measure of firms' exposure to state-dependent consumption risk. While [Zhang \(2005\)](#) demonstrates that firms with inflexible production technologies earn higher returns due to value risk, and [Novy-Marx \(2011\)](#) shows that operating leverage predicts returns through its connection to systematic risk, we document that the stability of operating leverage itself captures distinct variation in consumption risk exposure. Our findings extend [Cooper and Priestley \(2009\)](#) who examine time-varying risk premia by showing that cross-sectional differences in earnings leverage stability generate persistent return differentials through heterogeneous exposure to consumption disasters and state-dependent marginal utility. We advance the methodology for measuring consumption risk exposure beyond traditional approaches that rely on consumption betas or direct estimation of stochastic discount factors ([Lettau and Ludvigson, 2001](#)). By focusing on the stability of firms' operating characteristics, we provide a production-based measure that captures exposure to long-run consumption risks without requiring high-frequency consumption data or assumptions about utility functions. The ELS signal's ability to generate significant alphas relative to existing factors while maintaining distinct predictive power from related anomalies suggests it captures a fundamental dimension of systematic risk. Our evidence that ELS predicts returns across all size quintiles and remains robust to transaction costs has important implications for both academic research and practical portfolio management. The results suggest that investors require compensation for bearing earnings leverage instability risk because such firms provide poor consumption hedges during economic downturns when marginal utility is highest, consistent with the central predictions of consumption-based asset pricing theory.

## 2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of debt issuance activities and operating performance across firms. construction of our Earnings Leverage Stability signal relies on two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt instruments during the fiscal year, capturing new borrowing activities that extend the firm’s long-term capital structure. EBITDA (EBITDA) measures earnings before interest, taxes, depreciation, and amortization, providing a standardized metric of operating performance that excludes the effects of financing decisions, tax environments, and accounting choices related to capital investments. This measure serves as a proxy for the firm’s cash-generating ability from core operations. construct the Earnings Leverage Stability signal by computing the year-over-year change in long-term debt issuance scaled by lagged EBITDA, specifically calculated as  $(DLTIS(t) - DLTIS(t-1)) / EBITDA(t-1)$ . This ratio captures the change in a firm’s debt issuance activity relative to its operating cash generation capacity. A higher value indicates that a firm is accelerating its long-term borrowing relative to its earnings capacity, while a lower or negative value suggests either stable or declining reliance on debt financing relative to operating performance. We use annual observations based on fiscal year-end values and require non-missing values for both DLTIS and EBITDA variables. To ensure meaningful scaling, we exclude observations where lagged EBITDA is zero or negative, as these cases would produce economically uninterpretable ratios.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELS signal. Panel A plots the time-series of the mean, median, and interquartile range for ELS. On average, the cross-sectional mean (median) ELS is 0.03 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input ELS data. The signal's interquartile range spans -0.50 to 0.64. Panel B of Figure 1 plots the time-series of the coverage of the ELS signal for the CRSP universe. On average, the ELS signal is available for 6.21% of CRSP names, which on average make up 7.38% of total market capitalization.

### 4 Does ELS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELS portfolio and sells the low ELS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ELS strategy earns an average return of 0.25% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.27% to 0.30% per month and have t-statistics exceeding 3.66 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.06,

with a t-statistic of -3.49 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,232 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 4.37. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective

bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -7-19bps/month. The lowest return, (-7 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.26. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the ELS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELS, as well as average returns and alphas for long/short trading ELS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the ELS strategy achieves an average return of 25 bps/month with a t-statistic of 2.86. Among these large cap stocks, the alphas for the ELS strategy relative to the five most common factor models range from 23 to 31 bps/month with t-statistics between 2.54 and 3.47.

## 5 How does ELS perform relative to the zoo?

Figure 2 puts the performance of ELS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the ELS strategy falls in the distribution. The ELS strategy’s gross (net)

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.



Sharpe ratio of 0.46 (0.34) is greater than 87% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELS strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the ELS strategy would have yielded \$3.09 which ranks the ELS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELS strategy would have yielded \$1.79 which ranks the ELS strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELS relative to those. Panel A shows that the ELS strategy gross alphas fall between the 55 and 78 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELS ranks between the 70 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

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<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

## 6 Does ELS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELS with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELS or at least to weaken the power ELS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ELS}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ELS}ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ELS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELS signal in these Fama-MacBeth regressions exceed 1.08, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on ELS is 0.67.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELS strategy earns alphas that range from 23-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.17, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELS trading strategy achieves an alpha of 22bps/month with a t-statistic of 3.06.

## 7 Does ELS add relative to the whole zoo?

Finally, we can ask how much adding ELS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the ELS signal.<sup>4</sup> We consider

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<sup>4</sup>We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes ELS grows to \$1127.79.

## 8 Conclusion

This study provides compelling evidence that Earnings Leverage Stability (ELS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ELS-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of ELS as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.46 gross (0.34 net of transaction costs), indicating strong performance that remains economically meaningful after implementation costs. More notably, the strategy delivers statistically significant abnormal returns of 28 basis points per month (t-statistic = 3.68) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of a 22 basis point monthly alpha (t-statistic = 3.06) even after controlling for six closely related strategies from the factor zoo, including net debt financing, changes in financial liabilities, and accruals-based measures.

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ization on CRSP in the period for which ELS is available.

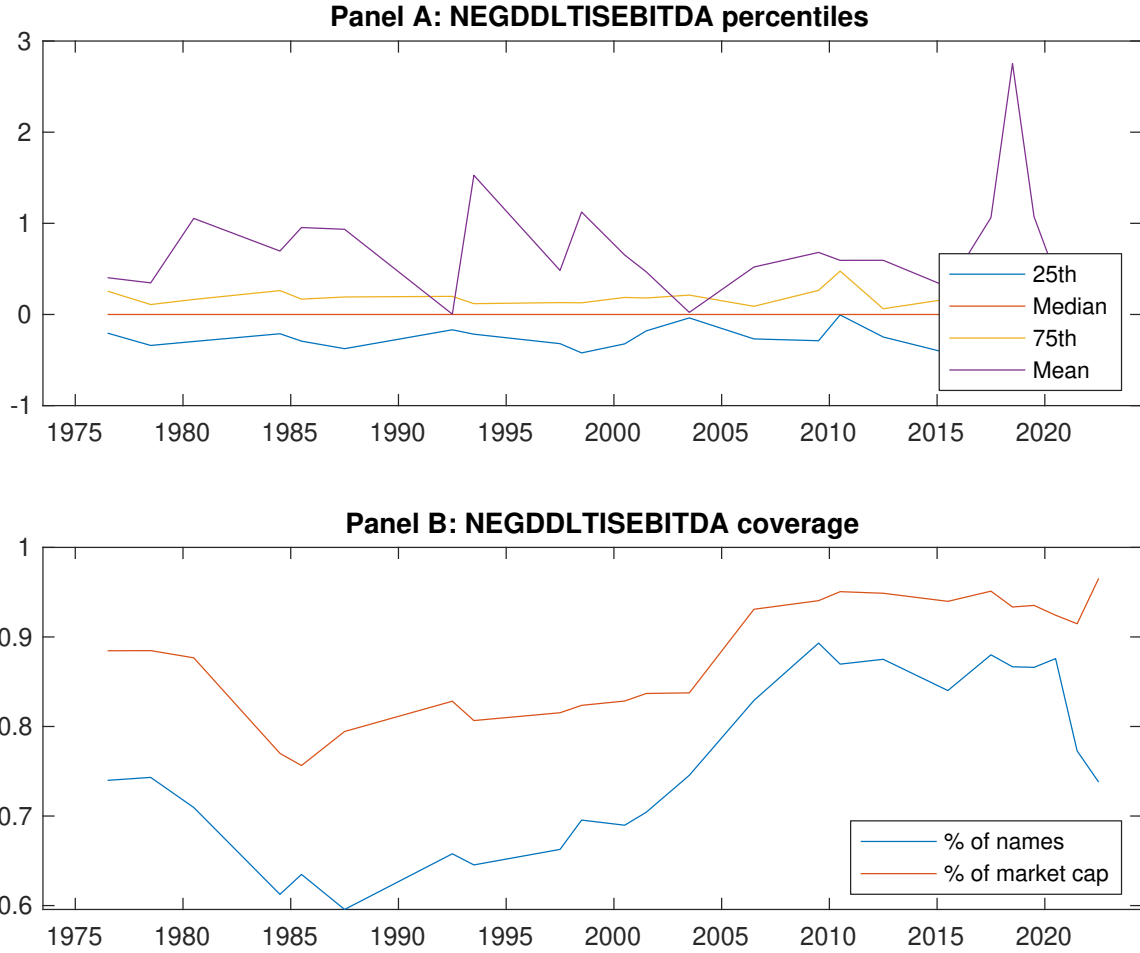
The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, ELS emerges as a distinct source of cross-sectional return predictability that cannot be fully explained by existing factor models or related accounting-based anomalies. For practitioners, the net-of-cost Sharpe ratio of 0.34 suggests that ELS-based strategies may offer economically viable investment opportunities, even after accounting for real-world implementation frictions.

Despite these promising results, several limitations warrant consideration. First, while our analysis accounts for transaction costs, other implementation challenges such as short-sale constraints and capital requirements may further impact real-world performance. Second, the stability of the ELS signal across different market regimes and international markets requires further investigation. Third, the economic mechanisms underlying the ELS anomaly remain incompletely understood, raising questions about whether the premium reflects systematic risk compensation or market inefficiency.

Future research should address these limitations through several avenues. Investigating the performance of ELS strategies in international equity markets would provide valuable out-of-sample validation and insights into the universality of the phenomenon. Additionally, exploring the economic drivers of the ELS premium through detailed fundamental analysis and connections to firm characteristics could enhance our theoretical understanding. Research examining the interaction between ELS and corporate financial policies, as well as its relationship to macroeconomic conditions, would further illuminate the sources of return predictability. Finally, investigating optimal portfolio construction techniques that incorporate ELS alongside other established factors could yield practical insights for multi-factor investment strategies.

In conclusion, Earnings Leverage Stability emerges as a robust and economically

significant predictor of equity returns that merits inclusion in the constellation of documented asset pricing anomalies. While questions remain about its underlying economic mechanisms and broader applicability, the evidence presented strongly supports its relevance for both academic asset pricing research and practical portfolio management.



**Figure 1:** Times series of ELS percentiles and coverage.  
This figure plots descriptive statistics for ELS. Panel A shows cross-sectional percentiles of ELS over the sample. Panel B plots the monthly coverage of ELS relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Excess returns and alphas on ELS-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.53<br>[2.43]   | 0.64<br>[3.55]   | 0.68<br>[3.46]   | 0.74<br>[4.24]   | 0.77<br>[3.73]   | 0.25<br>[3.33]   |
| $\alpha_{CAPM}$                                                                   | -0.21<br>[-3.52] | 0.03<br>[0.61]   | 0.02<br>[0.29]   | 0.15<br>[3.32]   | 0.07<br>[1.26]   | 0.28<br>[3.83]   |
| $\alpha_{FF3}$                                                                    | -0.24<br>[-4.21] | -0.01<br>[-0.31] | 0.07<br>[1.41]   | 0.14<br>[3.25]   | 0.06<br>[1.03]   | 0.30<br>[4.07]   |
| $\alpha_{FF4}$                                                                    | -0.21<br>[-3.61] | -0.00<br>[-0.09] | 0.11<br>[2.10]   | 0.11<br>[2.49]   | 0.06<br>[1.12]   | 0.27<br>[3.66]   |
| $\alpha_{FF5}$                                                                    | -0.22<br>[-3.80] | -0.06<br>[-1.49] | 0.10<br>[1.77]   | 0.06<br>[1.35]   | 0.07<br>[1.32]   | 0.30<br>[3.95]   |
| $\alpha_{FF6}$                                                                    | -0.20<br>[-3.39] | -0.05<br>[-1.24] | 0.12<br>[2.28]   | 0.04<br>[0.91]   | 0.08<br>[1.40]   | 0.28<br>[3.68]   |
| Panel B: Fama and French (2018) 6-factor model loadings for ELS-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 1.10<br>[80.87]  | 0.98<br>[100.50] | 0.98<br>[78.47]  | 0.96<br>[96.66]  | 1.04<br>[79.87]  | -0.06<br>[-3.49] |
| $\beta_{SMB}$                                                                     | 0.13<br>[6.46]   | -0.12<br>[-7.96] | -0.03<br>[-1.51] | -0.03<br>[-2.31] | 0.18<br>[9.03]   | 0.04<br>[1.70]   |
| $\beta_{HML}$                                                                     | 0.09<br>[3.43]   | 0.13<br>[7.05]   | -0.15<br>[-6.19] | -0.04<br>[-1.99] | -0.02<br>[-0.62] | -0.11<br>[-3.13] |
| $\beta_{RMW}$                                                                     | 0.04<br>[1.32]   | 0.11<br>[5.94]   | -0.01<br>[-0.31] | 0.11<br>[5.50]   | -0.05<br>[-2.02] | -0.09<br>[-2.54] |
| $\beta_{CMA}$                                                                     | -0.11<br>[-2.76] | 0.03<br>[1.12]   | -0.05<br>[-1.37] | 0.17<br>[5.75]   | 0.02<br>[0.66]   | 0.13<br>[2.64]   |
| $\beta_{UMD}$                                                                     | -0.04<br>[-2.67] | -0.02<br>[-1.61] | -0.04<br>[-3.51] | 0.03<br>[3.01]   | -0.01<br>[-0.61] | 0.03<br>[1.63]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 669              | 527              | 1047             | 577              | 656              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 1232             | 3166             | 2378             | 3251             | 1274             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

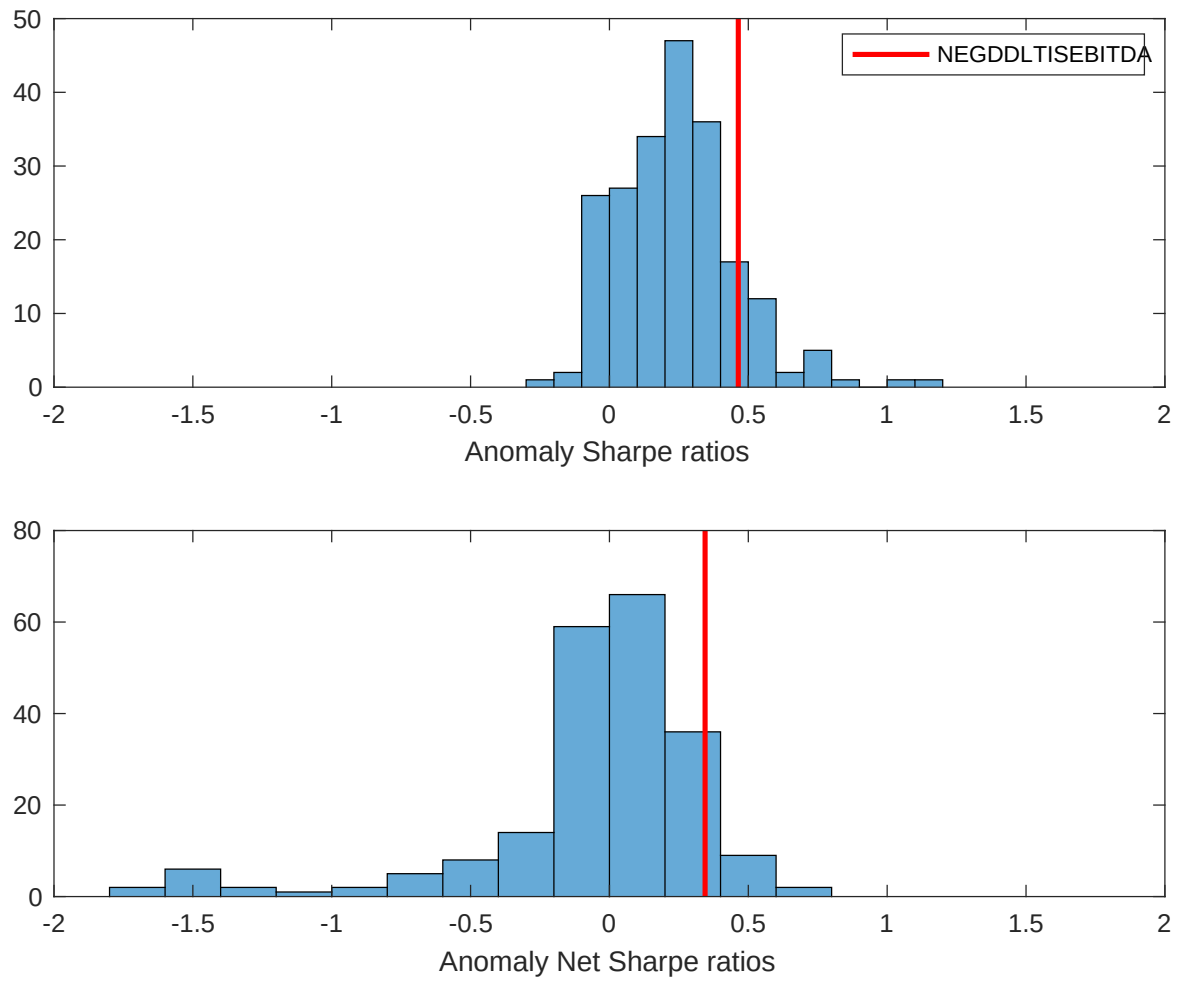
This table evaluates the robustness of the choices made in the ELS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.25<br>[3.33]     | 0.28<br>[3.83]           | 0.30<br>[4.07]          | 0.27<br>[3.66]          | 0.30<br>[3.95]          | 0.28<br>[3.68]          |
| Quintile                                                                 | NYSE   | EW      | 0.19<br>[4.37]     | 0.21<br>[4.87]           | 0.21<br>[4.84]          | 0.19<br>[4.43]          | 0.20<br>[4.75]          | 0.20<br>[4.51]          |
| Quintile                                                                 | Name   | VW      | 0.19<br>[2.61]     | 0.24<br>[3.23]           | 0.26<br>[3.56]          | 0.24<br>[3.24]          | 0.23<br>[3.13]          | 0.22<br>[2.97]          |
| Quintile                                                                 | Cap    | VW      | 0.22<br>[3.47]     | 0.26<br>[4.02]           | 0.28<br>[4.37]          | 0.24<br>[3.69]          | 0.21<br>[3.31]          | 0.19<br>[2.90]          |
| Decile                                                                   | NYSE   | VW      | 0.27<br>[2.83]     | 0.30<br>[3.14]           | 0.31<br>[3.27]          | 0.30<br>[3.16]          | 0.32<br>[3.30]          | 0.32<br>[3.25]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.18<br>[2.47]     | 0.22<br>[2.99]           | 0.23<br>[3.17]          | 0.22<br>[2.97]          | 0.23<br>[3.07]          | 0.21<br>[2.89]          |
| Quintile                                                                 | NYSE   | EW      | -0.07<br>[-1.26]   |                          |                         |                         |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.13<br>[1.75]     | 0.17<br>[2.39]           | 0.19<br>[2.65]          | 0.18<br>[2.49]          | 0.17<br>[2.34]          | 0.16<br>[2.19]          |
| Quintile                                                                 | Cap    | VW      | 0.17<br>[2.66]     | 0.21<br>[3.26]           | 0.22<br>[3.53]          | 0.20<br>[3.18]          | 0.17<br>[2.66]          | 0.15<br>[2.38]          |
| Decile                                                                   | NYSE   | VW      | 0.19<br>[1.99]     | 0.21<br>[2.20]           | 0.22<br>[2.32]          | 0.22<br>[2.29]          | 0.22<br>[2.27]          | 0.21<br>[2.22]          |

**Table 3:** Conditional sort on size and ELS

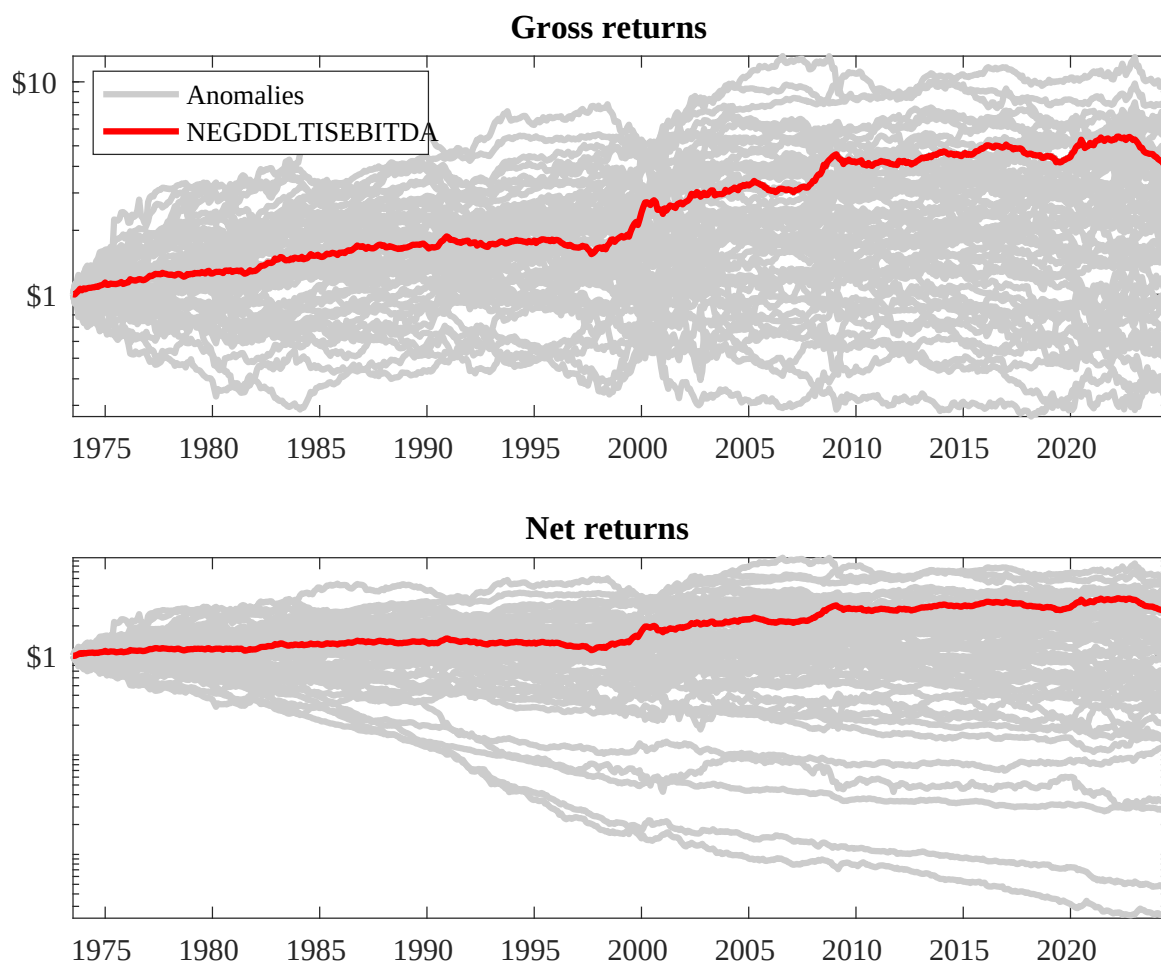
This table presents results for conditional double sorts on size and ELS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELS and short stocks with low ELS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                                                    |                 |                |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | ELS Quintiles |                |                |                |                | ELS Strategies                                     |                 |                |                |                |                |                |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |                |
|                                                                       | (1)           | 0.66<br>[2.44] | 0.90<br>[3.24] | 0.93<br>[3.43] | 0.93<br>[3.21] | 0.74<br>[2.73]                                     | 0.08<br>[1.03]  | 0.09<br>[1.16] | 0.08<br>[1.06] | 0.04<br>[0.53] | 0.05<br>[0.69] | 0.03<br>[0.35] |
|                                                                       | (2)           | 0.71<br>[2.69] | 0.84<br>[3.39] | 0.88<br>[3.47] | 0.89<br>[3.62] | 0.87<br>[3.38]                                     | 0.16<br>[2.22]  | 0.19<br>[2.54] | 0.16<br>[2.16] | 0.17<br>[2.25] | 0.17<br>[2.23] | 0.18<br>[2.31] |
|                                                                       | (3)           | 0.77<br>[3.10] | 0.84<br>[3.81] | 0.79<br>[3.30] | 0.84<br>[3.78] | 0.87<br>[3.64]                                     | 0.10<br>[1.22]  | 0.13<br>[1.69] | 0.13<br>[1.66] | 0.11<br>[1.41] | 0.15<br>[1.82] | 0.13<br>[1.62] |
|                                                                       | (4)           | 0.64<br>[2.78] | 0.84<br>[4.05] | 0.85<br>[3.86] | 0.75<br>[3.61] | 0.91<br>[4.16]                                     | 0.27<br>[3.67]  | 0.30<br>[4.04] | 0.32<br>[4.27] | 0.30<br>[3.93] | 0.34<br>[4.38] | 0.32<br>[4.13] |
|                                                                       | (5)           | 0.47<br>[2.32] | 0.59<br>[3.36] | 0.64<br>[3.27] | 0.69<br>[3.85] | 0.72<br>[3.71]                                     | 0.25<br>[2.86]  | 0.28<br>[3.14] | 0.31<br>[3.47] | 0.28<br>[3.13] | 0.24<br>[2.69] | 0.23<br>[2.54] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                                                    |                 |                |                |                |                |                |
| Size quintiles                                                        | ELS Quintiles |                |                |                |                | ELS Quintiles                                      |                 |                |                |                |                |                |
|                                                                       | Average $n$   |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |                |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |                |
|                                                                       | (1)           | 390            | 391            | 392            | 391            | 388                                                | 36              | 34             | 33             | 34             | 35             |                |
|                                                                       | (2)           | 106            | 107            | 106            | 107            | 106                                                | 60              | 61             | 60             | 61             | 61             |                |
|                                                                       | (3)           | 76             | 76             | 76             | 76             | 76                                                 | 107             | 109            | 104            | 107            | 106            |                |
|                                                                       | (4)           | 64             | 64             | 64             | 64             | 63                                                 | 226             | 240            | 228            | 236            | 226            |                |
| (5)                                                                   | 58            | 59             | 59             | 59             | 59             | 1322                                               | 2164            | 1961           | 2271           | 1420           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

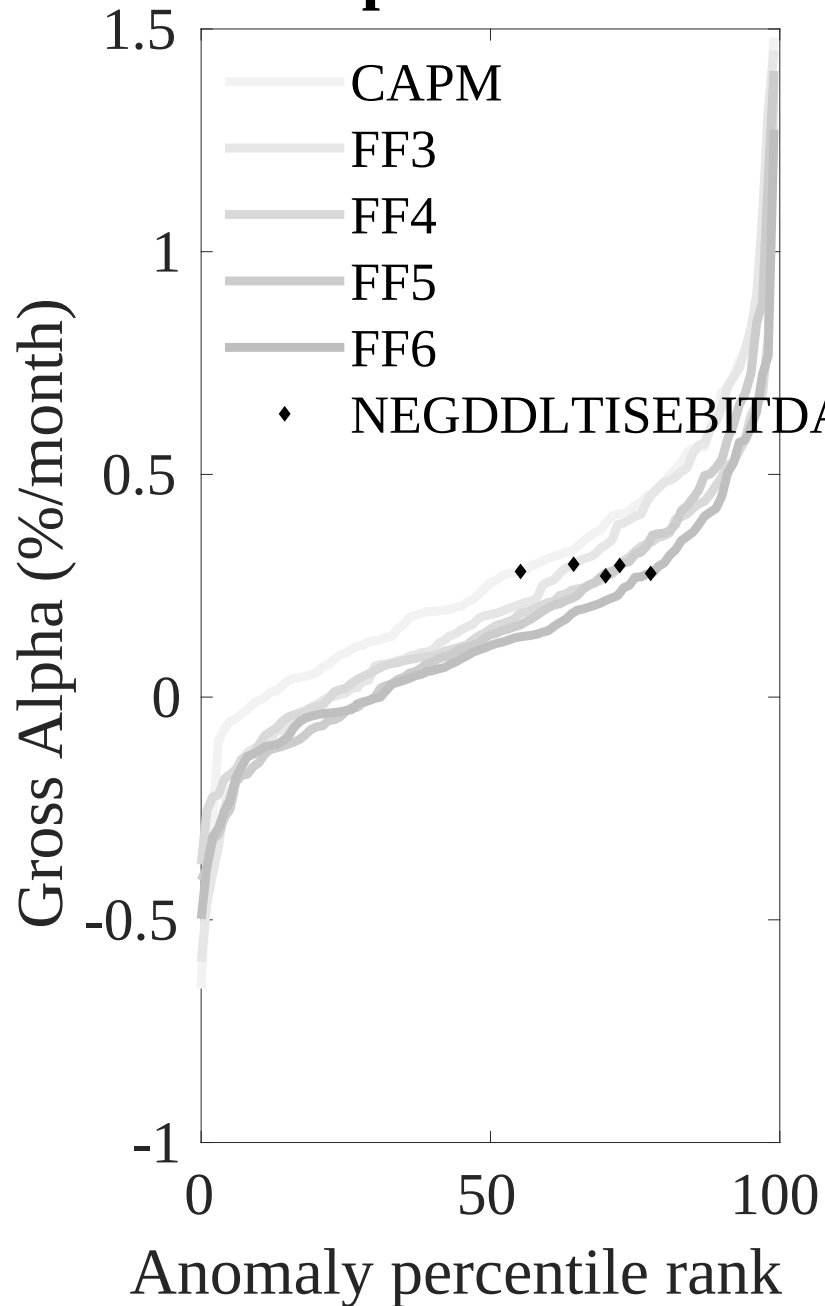
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

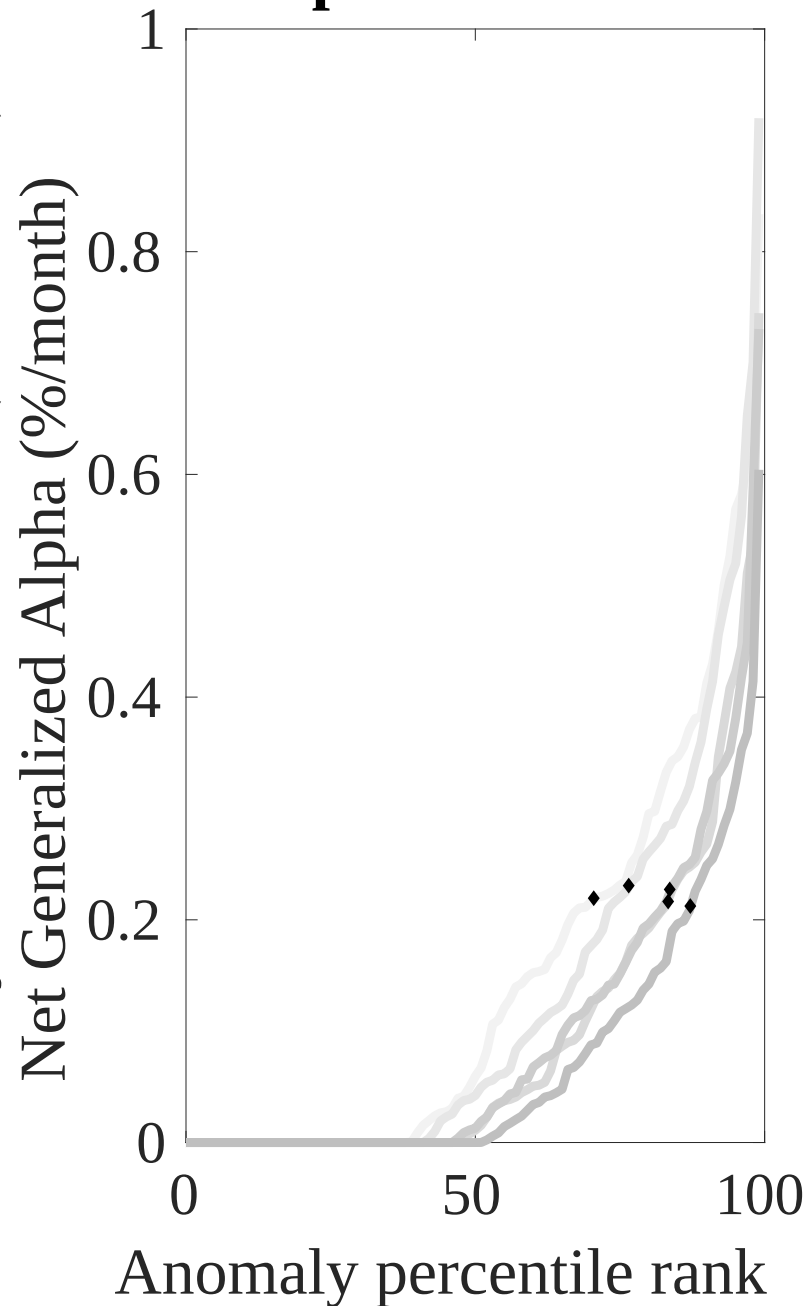
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

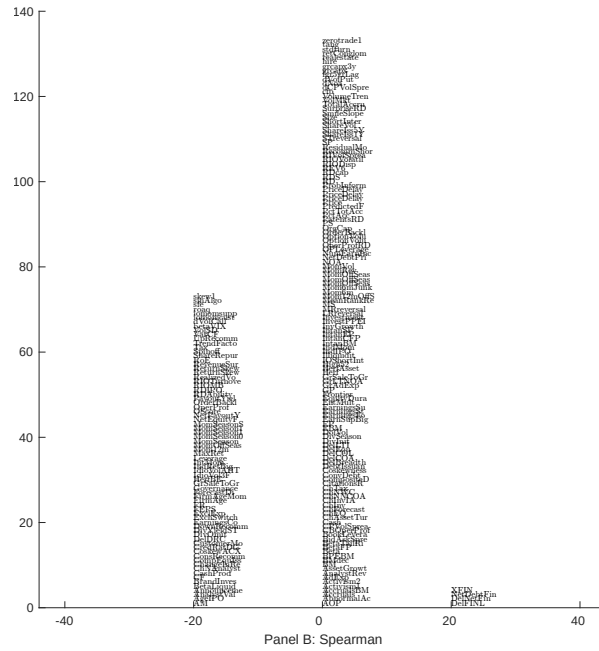
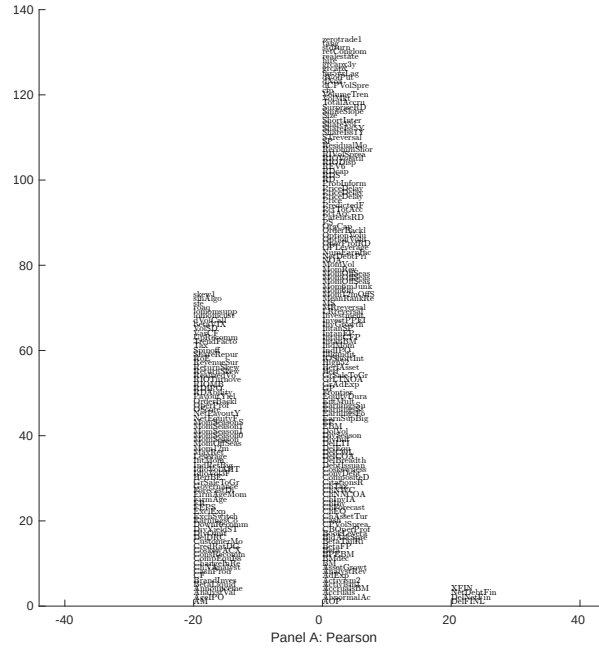
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

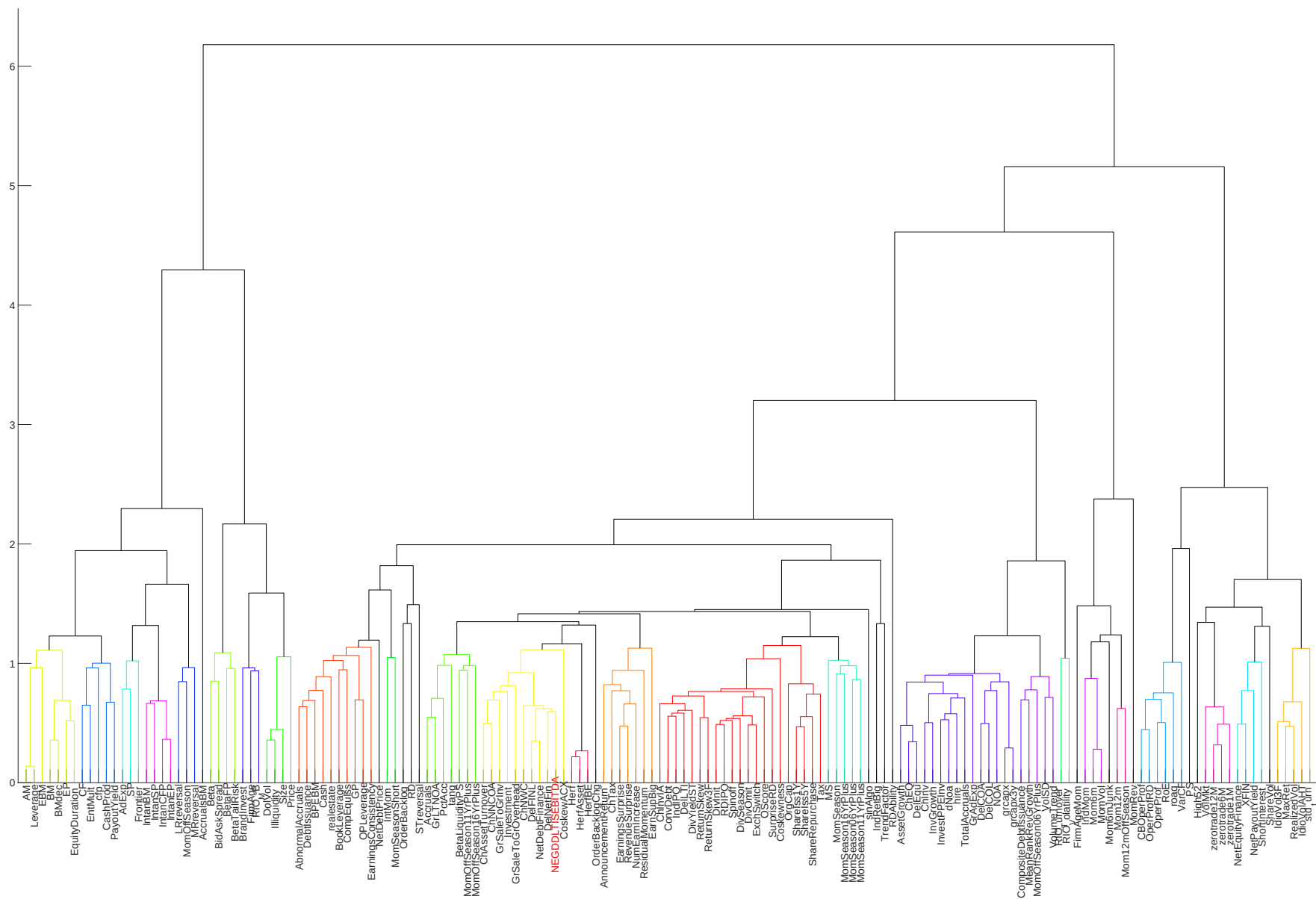
## Net Alpha Percentiles





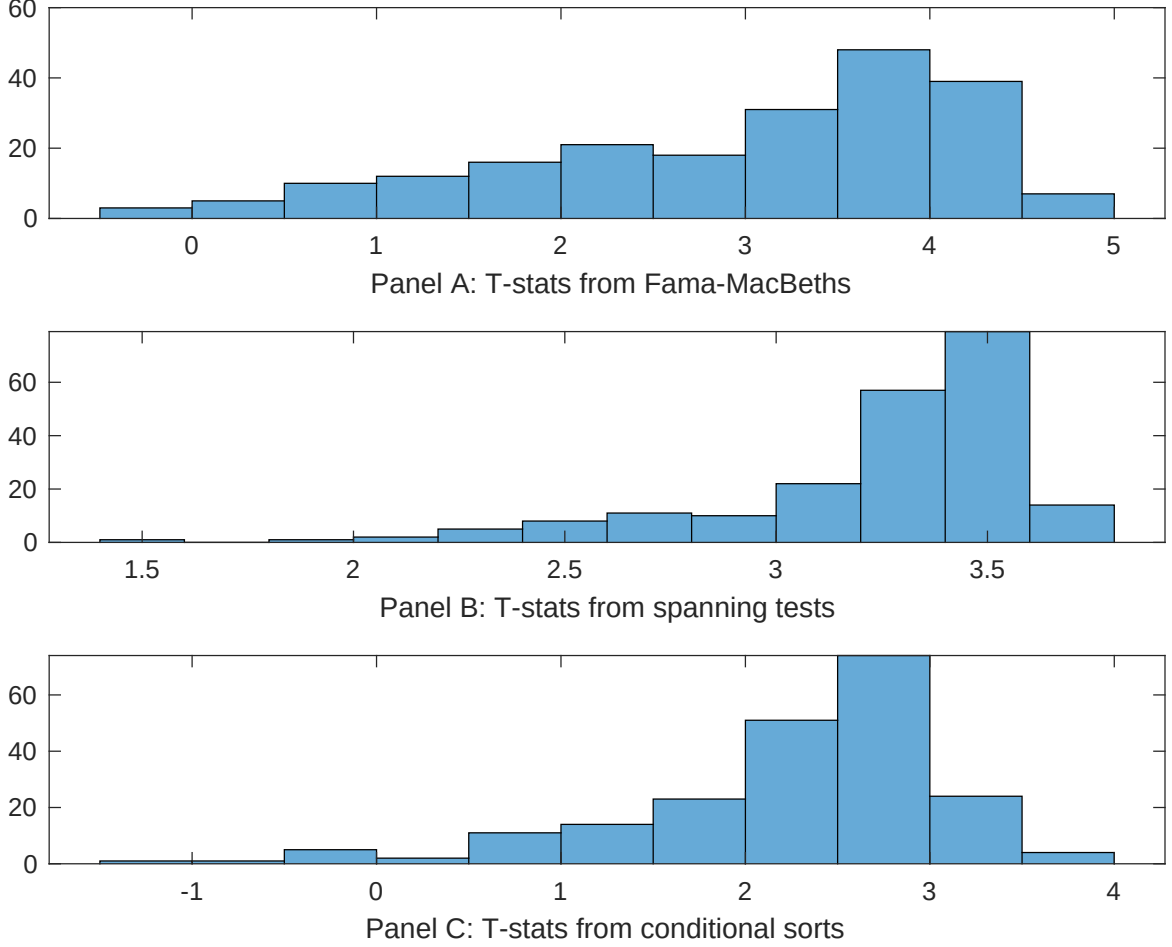
**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ELS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ELS}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 210 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

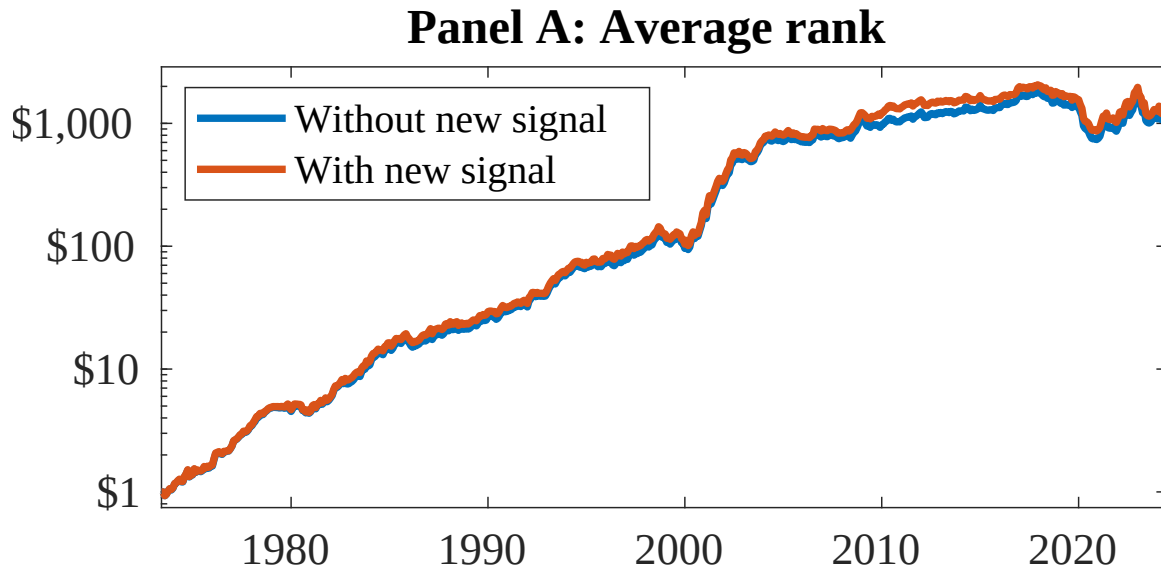
This table presents Fama-MacBeth results of returns on ELS. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.13<br>[5.32] | 0.13<br>[5.36] | 0.13<br>[5.22] | 0.12<br>[4.99] | 0.13<br>[5.29] | 0.14<br>[5.96] | 0.13<br>[5.64]   |
| ELS             | 0.91<br>[1.21] | 0.79<br>[1.08] | 0.18<br>[2.39] | 0.24<br>[3.27] | 0.30<br>[3.57] | 0.17<br>[1.89] | 0.72<br>[0.67]   |
| Anomaly 1       | 0.20<br>[9.24] |                |                |                |                |                | 0.21<br>[3.94]   |
| Anomaly 2       |                | 0.18<br>[9.52] |                |                |                |                | -0.13<br>[-0.22] |
| Anomaly 3       |                |                | 0.73<br>[4.92] |                |                |                | -0.68<br>[-1.76] |
| Anomaly 4       |                |                |                | 0.14<br>[4.41] |                |                | 0.91<br>[1.94]   |
| Anomaly 5       |                |                |                |                | 0.42<br>[7.24] |                | 0.23<br>[2.91]   |
| Anomaly 6       |                |                |                |                |                | 0.11<br>[5.84] | 0.53<br>[2.86]   |
| # months        | 612            | 612            | 612            | 612            | 612            | 612            | 612              |
| $\bar{R}^2(\%)$ | 0              | 0              | 0              | 0              | 0              | 0              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{ELS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

|                 |                   |                   |                  |                  |                  |                   |                  |
|-----------------|-------------------|-------------------|------------------|------------------|------------------|-------------------|------------------|
| Intercept       | 0.25<br>[3.50]    | 0.25<br>[3.41]    | 0.23<br>[3.17]   | 0.24<br>[3.25]   | 0.28<br>[3.72]   | 0.28<br>[3.76]    | 0.22<br>[3.06]   |
| Anomaly 1       | 24.19<br>[6.09]   |                   |                  |                  |                  |                   | 17.44<br>[3.37]  |
| Anomaly 2       |                   | 21.67<br>[5.22]   |                  |                  |                  |                   | -1.95<br>[-0.33] |
| Anomaly 3       |                   |                   | 20.17<br>[5.84]  |                  |                  |                   | 12.64<br>[3.27]  |
| Anomaly 4       |                   |                   |                  | 10.64<br>[3.62]  |                  |                   | 7.43<br>[2.56]   |
| Anomaly 5       |                   |                   |                  |                  | 5.85<br>[2.02]   |                   | 5.01<br>[1.80]   |
| Anomaly 6       |                   |                   |                  |                  |                  | 10.70<br>[3.12]   | 6.46<br>[1.86]   |
| mkt             | -5.79<br>[-3.47]  | -5.34<br>[-3.17]  | -5.46<br>[-3.26] | -5.03<br>[-2.93] | -5.96<br>[-3.47] | -5.27<br>[-3.07]  | -4.75<br>[-2.86] |
| smb             | 3.34<br>[1.31]    | 3.26<br>[1.26]    | 6.65<br>[2.62]   | 7.50<br>[2.85]   | 6.13<br>[2.34]   | 3.66<br>[1.38]    | 5.72<br>[2.14]   |
| hml             | -8.65<br>[-2.68]  | -8.19<br>[-2.51]  | -9.98<br>[-3.09] | -6.93<br>[-2.06] | -9.80<br>[-2.96] | -11.06<br>[-3.33] | -8.49<br>[-2.58] |
| rmw             | -11.12<br>[-3.37] | -10.42<br>[-3.14] | -6.05<br>[-1.83] | -4.97<br>[-1.43] | -7.62<br>[-2.25] | -9.49<br>[-2.82]  | -6.14<br>[-1.77] |
| cma             | 6.89<br>[1.38]    | 5.92<br>[1.15]    | 18.32<br>[3.67]  | 9.74<br>[1.92]   | 7.95<br>[1.41]   | 10.26<br>[2.02]   | 4.01<br>[0.68]   |
| umd             | 1.45<br>[0.86]    | 1.13<br>[0.66]    | 2.66<br>[1.60]   | 2.20<br>[1.29]   | 2.41<br>[1.39]   | 2.57<br>[1.51]    | 0.47<br>[0.28]   |
| # months        | 612               | 612               | 612              | 612              | 612              | 612               | 612              |
| $\bar{R}^2(\%)$ | 12                | 10                | 11               | 8                | 7                | 8                 | 15               |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as ELS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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